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## Pin Map

| Pin Name | Pin | Pin Name | Pin |
|----------|-----|----------|-----|
| 1CP      | 1   | 1R       | 2   |
| 1K       | 3   | 2CP      | 5   |
| 2R       | 6   | 2J       | 7   |
| 2Qnot    | 14  | 2Q       | 15  |
| 2K       | 16  | 1Q       | 18  |
| 1Qnot    | 19  | 1J       | 20  |

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## Global Parameters

| Parameter   | Value                      |
|-------------|----------------------------|
| VCC Pin     | 4                          |
| GND Pin     | 17                         |
| VCC Voltage | 2.5, 3.3, 4.0, 4.5, 5      |
| Input Low   | 0.75, 0.99, 1.2, 1.35, 1.5 |
| Input High  | 1.75, 2.31, 2.8, 3.15, 3.5 |
| Output Low  | 0.3, 0.3, 0.3, 0.3, 0.3    |
| Output High | 2.2, 3.0, 3.6, 4.1, 4.7    |

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## Tests: IC PASS 7/7

### JK Flip-Flop 1: PASS

| Inputs |    |    |     |                 | VCC | Outputs/Results |         |       |          |
|--------|----|----|-----|-----------------|-----|-----------------|---------|-------|----------|
| 1R     | 1J | 1K | 1CP | 2R, 2J, 2K, 2CP |     | 1Q              |         | 1Qnot |          |
| L      | X  | X  | X   | L               | 2.5 | L               | 0.0 (L) | H     | 2.37 (H) |
|        |    |    |     |                 | 3.3 |                 | 0.0 (L) |       | 3.15 (H) |
|        |    |    |     |                 | 4.0 |                 | 0.0 (L) |       | 3.81 (H) |
|        |    |    |     |                 | 4.5 |                 | 0.0 (L) |       | 4.43 (H) |
|        |    |    |     |                 | 5   |                 | 0.0 (L) |       | 4.85 (H) |

| Inputs |    |    |     |                 | VCC | Outputs/Results |          |       |          |
|--------|----|----|-----|-----------------|-----|-----------------|----------|-------|----------|
| 1R     | 1J | 1K | 1CP | 2R, 2J, 2K, 2CP |     | 1Q              |          | 1Qnot |          |
| H      | H  | L  | F   | L               | 2.5 | H               | 2.37 (H) | L     | 0.0 (L)  |
|        |    |    |     |                 | 3.3 |                 | 3.15 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 4.0 |                 | 3.81 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 4.5 |                 | 4.43 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 5   |                 | 4.86 (H) |       | 0.0 (L)  |
| H      | L  | H  | F   | L               | 2.5 | L               | 0.0 (L)  | H     | 2.37 (H) |
|        |    |    |     |                 | 3.3 |                 | 0.0 (L)  |       | 3.15 (H) |
|        |    |    |     |                 | 4.0 |                 | 0.0 (L)  |       | 3.81 (H) |
|        |    |    |     |                 | 4.5 |                 | 0.0 (L)  |       | 4.43 (H) |
|        |    |    |     |                 | 5   |                 | 0.0 (L)  |       | 4.86 (H) |

### JK Flip-Flop 2: **PASS**

| Inputs |    |    |     |                 | VCC | Outputs/Results |         |       |          |
|--------|----|----|-----|-----------------|-----|-----------------|---------|-------|----------|
| 2R     | 2J | 2K | 2CP | 1R, 1J, 1K, 1CP |     | 2Q              |         | 2Qnot |          |
| L      | X  | X  | X   | L               | 2.5 | L               | 0.0 (L) | H     | 2.37 (H) |
|        |    |    |     |                 | 3.3 |                 | 0.0 (L) |       | 3.15 (H) |
|        |    |    |     |                 | 4.0 |                 | 0.0 (L) |       | 3.81 (H) |
|        |    |    |     |                 | 4.5 |                 | 0.0 (L) |       | 4.43 (H) |
|        |    |    |     |                 | 5   |                 | 0.0 (L) |       | 4.86 (H) |

| Inputs |    |    |     |                 | VCC | Outputs/Results |          |       |          |
|--------|----|----|-----|-----------------|-----|-----------------|----------|-------|----------|
| 2R     | 2J | 2K | 2CP | 1R, 1J, 1K, 1CP |     | 2Q              |          | 2Qnot |          |
| H      | H  | L  | F   | L               | 2.5 | H               | 2.37 (H) | L     | 0.0 (L)  |
|        |    |    |     |                 | 3.3 |                 | 3.15 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 4.0 |                 | 3.81 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 4.5 |                 | 4.43 (H) |       | 0.0 (L)  |
|        |    |    |     |                 | 5   |                 | 4.86 (H) |       | 0.0 (L)  |
| H      | L  | H  | F   | L               | 2.5 | L               | 0.0 (L)  | H     | 2.37 (H) |
|        |    |    |     |                 | 3.3 |                 | 0.0 (L)  |       | 3.15 (H) |
|        |    |    |     |                 | 4.0 |                 | 0.0 (L)  |       | 3.81 (H) |
|        |    |    |     |                 | 4.5 |                 | 0.0 (L)  |       | 4.43 (H) |
|        |    |    |     |                 | 5   |                 | 0.0 (L)  |       | 4.85 (H) |

**No Change 1 -> L: PASS**

| Inputs |        |        |          | VCC | Outputs/Results |         |    |         |       |          |       |          |
|--------|--------|--------|----------|-----|-----------------|---------|----|---------|-------|----------|-------|----------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |         | 2Q |         | 1Qnot |          | 2Qnot |          |
| L      | X      | X      | X        | 2.5 | L               | 0.0 (L) | L  | 0.0 (L) | H     | 2.37 (H) | H     | 2.37 (H) |
|        |        |        |          | 3.3 |                 | 0.0 (L) |    | 0.0 (L) |       | 3.15 (H) |       | 3.15 (H) |
|        |        |        |          | 4.0 |                 | 0.0 (L) |    | 0.0 (L) |       | 3.81 (H) |       | 3.81 (H) |
|        |        |        |          | 4.5 |                 | 0.0 (L) |    | 0.0 (L) |       | 4.43 (H) |       | 4.43 (H) |
|        |        |        |          | 5   |                 | 0.0 (L) |    | 0.0 (L) |       | 4.86 (H) |       | 4.86 (H) |

| Inputs |        |        |          | VCC | Outputs/Results |         |    |         |       |          |       |          |
|--------|--------|--------|----------|-----|-----------------|---------|----|---------|-------|----------|-------|----------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |         | 2Q |         | 1Qnot |          | 2Qnot |          |
| H      | L      | L      | F        | 2.5 | L               | 0.0 (L) | L  | 0.0 (L) | H     | 2.37 (H) | H     | 2.37 (H) |
|        |        |        |          | 3.3 |                 | 0.0 (L) |    | 0.0 (L) |       | 3.15 (H) |       | 3.15 (H) |
|        |        |        |          | 4.0 |                 | 0.0 (L) |    | 0.0 (L) |       | 3.81 (H) |       | 3.81 (H) |
|        |        |        |          | 4.5 |                 | 0.0 (L) |    | 0.0 (L) |       | 4.43 (H) |       | 4.43 (H) |
|        |        |        |          | 5   |                 | 0.0 (L) |    | 0.0 (L) |       | 4.86 (H) |       | 4.86 (H) |

**No Change 1 -> H: PASS**

| Inputs |        |        |          | VCC | Outputs/Results |          |    |          |       |         |       |         |
|--------|--------|--------|----------|-----|-----------------|----------|----|----------|-------|---------|-------|---------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |          | 2Q |          | 1Qnot |         | 2Qnot |         |
| H      | H      | L      | F        | 2.5 | H               | 2.37 (H) | H  | 2.37 (H) | L     | 0.0 (L) | L     | 0.0 (L) |
|        |        |        |          | 3.3 |                 | 3.15 (H) |    | 3.15 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 4.0 |                 | 3.81 (H) |    | 3.81 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 4.5 |                 | 4.43 (H) |    | 4.43 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 5   |                 | 4.86 (H) |    | 4.86 (H) |       | 0.0 (L) |       | 0.0 (L) |
| H      | L      | L      | F        | 2.5 | H               | 2.37 (H) | H  | 2.37 (H) | L     | 0.0 (L) | L     | 0.0 (L) |
|        |        |        |          | 3.3 |                 | 3.15 (H) |    | 3.15 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 4.0 |                 | 3.81 (H) |    | 3.81 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 4.5 |                 | 4.43 (H) |    | 4.43 (H) |       | 0.0 (L) |       | 0.0 (L) |
|        |        |        |          | 5   |                 | 4.86 (H) |    | 4.86 (H) |       | 0.0 (L) |       | 0.0 (L) |

**No Change 2 -> L: PASS**

| Inputs |        |        |          | VCC | Outputs/Results |            |    |            |       |             |       |             |
|--------|--------|--------|----------|-----|-----------------|------------|----|------------|-------|-------------|-------|-------------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |            | 2Q |            | 1Qnot |             | 2Qnot |             |
| L      | X      | X      | X        | 2.5 | L               | 0.0<br>(L) | L  | 0.0<br>(L) | H     | 2.37<br>(H) | H     | 2.37<br>(H) |
|        |        |        |          | 3.3 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.15<br>(H) |       | 3.15<br>(H) |
|        |        |        |          | 4.0 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.81<br>(H) |       | 3.81<br>(H) |
|        |        |        |          | 4.5 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.44<br>(H) |       | 4.44<br>(H) |
|        |        |        |          | 5   |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.86<br>(H) |       | 4.86<br>(H) |
| H      | X      | X      | H        | 2.5 | L               | 0.0<br>(L) | L  | 0.0<br>(L) | H     | 2.37<br>(H) | H     | 2.37<br>(H) |
|        |        |        |          | 3.3 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.15<br>(H) |       | 3.15<br>(H) |
|        |        |        |          | 4.0 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.81<br>(H) |       | 3.81<br>(H) |
|        |        |        |          | 4.5 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.44<br>(H) |       | 4.44<br>(H) |
|        |        |        |          | 5   |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.86<br>(H) |       | 4.86<br>(H) |

**No Change 2 -> H: PASS**

| Inputs |        |        |          | VCC | Outputs/Results |             |    |             |       |            |       |            |
|--------|--------|--------|----------|-----|-----------------|-------------|----|-------------|-------|------------|-------|------------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |             | 2Q |             | 1Qnot |            | 2Qnot |            |
| H      | H      | L      | F        | 2.5 | H               | 2.37<br>(H) | H  | 2.37<br>(H) | L     | 0.0<br>(L) | L     | 0.0<br>(L) |
|        |        |        |          | 3.3 |                 | 3.15<br>(H) |    | 3.15<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 4.0 |                 | 3.81<br>(H) |    | 3.81<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 4.5 |                 | 4.43<br>(H) |    | 4.43<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 5   |                 | 4.86<br>(H) |    | 4.86<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |

| Inputs |        |        |          | VCC | Outputs/Results |             |    |             |       |            |       |            |
|--------|--------|--------|----------|-----|-----------------|-------------|----|-------------|-------|------------|-------|------------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |             | 2Q |             | 1Qnot |            | 2Qnot |            |
| H      | X      | X      | H        | 2.5 | H               | 2.37<br>(H) | H  | 2.37<br>(H) | L     | 0.0<br>(L) | L     | 0.0<br>(L) |
|        |        |        |          | 3.3 |                 | 3.15<br>(H) |    | 3.15<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 4.0 |                 | 3.81<br>(H) |    | 3.81<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 4.5 |                 | 4.43<br>(H) |    | 4.43<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |
|        |        |        |          | 5   |                 | 4.86<br>(H) |    | 4.86<br>(H) |       | 0.0<br>(L) |       | 0.0<br>(L) |

**Toggle: *PASS***

| Inputs |        |        |          | VCC | Outputs/Results |             |    |             |       |             |       |             |
|--------|--------|--------|----------|-----|-----------------|-------------|----|-------------|-------|-------------|-------|-------------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |             | 2Q |             | 1Qnot |             | 2Qnot |             |
| L      | X      | X      | X        | 2.5 | L               | 0.0<br>(L)  | L  | 0.0<br>(L)  | H     | 2.37<br>(H) | H     | 2.37<br>(H) |
|        |        |        |          | 3.3 |                 | 0.0<br>(L)  |    | 0.0<br>(L)  |       | 3.15<br>(H) |       | 3.15<br>(H) |
|        |        |        |          | 4.0 |                 | 0.0<br>(L)  |    | 0.0<br>(L)  |       | 3.81<br>(H) |       | 3.81<br>(H) |
|        |        |        |          | 4.5 |                 | 0.0<br>(L)  |    | 0.0<br>(L)  |       | 4.44<br>(H) |       | 4.43<br>(H) |
|        |        |        |          | 5   |                 | 0.0<br>(L)  |    | 0.0<br>(L)  |       | 4.86<br>(H) |       | 4.85<br>(H) |
| H      | H      | H      | F        | 2.5 | H               | 2.37<br>(H) | H  | 2.37<br>(H) | L     | 0.0<br>(L)  | L     | 0.0<br>(L)  |
|        |        |        |          | 3.3 |                 | 3.15<br>(H) |    | 3.15<br>(H) |       | 0.0<br>(L)  |       | 0.0<br>(L)  |
|        |        |        |          | 4.0 |                 | 3.81<br>(H) |    | 3.81<br>(H) |       | 0.0<br>(L)  |       | 0.0<br>(L)  |
|        |        |        |          | 4.5 |                 | 4.43<br>(H) |    | 4.43<br>(H) |       | 0.0<br>(L)  |       | 0.0<br>(L)  |
|        |        |        |          | 5   |                 | 4.85<br>(H) |    | 4.85<br>(H) |       | 0.0<br>(L)  |       | 0.0<br>(L)  |

| Inputs |        |        |          | VCC | Outputs/Results |            |    |            |       |             |       |             |
|--------|--------|--------|----------|-----|-----------------|------------|----|------------|-------|-------------|-------|-------------|
| 1R, 2R | 1J, 2J | 1K, 2K | 1CP, 2CP |     | 1Q              |            | 2Q |            | 1Qnot |             | 2Qnot |             |
| H      | H      | H      | F        | 2.5 | L               | 0.0<br>(L) | L  | 0.0<br>(L) | H     | 2.37<br>(H) | H     | 2.37<br>(H) |
|        |        |        |          | 3.3 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.15<br>(H) |       | 3.15<br>(H) |
|        |        |        |          | 4.0 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 3.81<br>(H) |       | 3.81<br>(H) |
|        |        |        |          | 4.5 |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.43<br>(H) |       | 4.43<br>(H) |
|        |        |        |          | 5   |                 | 0.0<br>(L) |    | 0.0<br>(L) |       | 4.86<br>(H) |       | 4.86<br>(H) |